

Conjecture 00000003968: the graceful-labeling counts of caterpillars have no rational generating function

Statement and reading

Conjecture. The generating function of the number of graceful labelings of n -vertex caterpillar trees is a rational function, with denominator given by the partition function of the caterpillar spine length.

Verdict: disproved. We prove that the ordinary generating function $\sum_n a_n z^n$ is not rational, for four counting conventions a_n . The clause about the denominator (“partition function of the spine length”) has no precise meaning and is not formalized; refuting rationality refutes the conjunction.

Definitions (Lean: C3968.IsCaterpillar, IsGraceful). A *graceful labeling* of a graph G with n vertices is a bijection $f : V \rightarrow \{0, \dots, n-1\}$ such that for each $d \in \{1, \dots, n-1\}$ exactly one edge $\{u, v\}$ has $|f(u) - f(v)| = d$. A *caterpillar* is a tree with a path S (an induced subgraph isomorphic to the path graph on k vertices) such that every vertex lies on S or is adjacent to a vertex of S (equivalently, deleting all leaves leaves a path). *Rational* (C3968.Rational) means $Q \cdot A = P$ in $\mathbb{Q}[[z]]$ for polynomials P, Q with $Q \neq 0$.

Counting conventions refuted (all in Lean). (A) **classCount**: sum over isomorphism classes of caterpillars on $\{0..n-1\}$ (one representative each) of the number of graceful bijections. (B) **pairCount**: pairs (caterpillar on $\{0..n-1\}$, graceful bijection). (C) **edgeSetCount**: labelings up to automorphism, i.e. graceful caterpillar edge sets on the label set $\{0..n-1\}$. (D) **complementCount**: as (C), with an edge set identified with its complement $x \mapsto n-1-x$. *Not refuted*: the exponential generating function; (A) and (B) with a labeling identified with its complement (this at most halves the count, so the same argument applies, but it is not in Lean).

Proof

Lemma 1 (exponential bound; rational_bound). *If $QA = P$ with $Q \neq 0$, then $a_n \leq MR^n$ for some $M > 0, R \geq 1$.*

Proof. Write $Q = X^k Q_1$ with $Q_1(0) \neq 0$. For $n > \deg P$ the coefficient of z^{n+k} gives $\sum_{i+j=n} q_i a_j = 0$, so $a_{m+1} = -\sum_{i+j=m} (q_{i+1}/q_0) a_j$ for $m+1 > \deg P + 1$. Put $c = \sum_i |q_{i+1}/q_0|$, $R = c + 1$, and $M = 1 + \sum_{j \leq \deg P+2} a_j$. Strong induction gives $a_{m+1} \leq cMR^m \leq MR^{m+1}$. $\square$

Lemma 2 (family; fam, fam_tree, fam_cat, fam_graceful). *Let $s \geq 1$, $n = 12s$ and τ a permutation of $\{0..s-1\}$. Define a parent map on $\{0..n-1\}$: for $v < 4s$, $v \neq 0$: $\text{par}(v) = n - v$; $\text{par}(0) = 0$; for $4s \leq v < 7s$: $4s - 1$; for $7s \leq v < 8s$: $v + 3s + 1 + \tau(v - 7s)$; for $v \geq 8s$: $n - 1 - v$. The graph G_τ has edges $\{v, \text{par}(v)\}$, $v \neq 0$. It is a caterpillar with graceful identity labeling.*

Proof. *Tree* (`tree_of_par`): the rank $2v$ (low $v < 4s$), $n-1-v \mapsto 2(n-1-v)+1$ (high), $n+v$ (leaf) strictly decreases along `par` and 0 is the root, so G_τ is connected with $n-1$ edges. *Caterpillar*: the vertices $< 4s$ or $\geq 8s$ form the zigzag path $0, n-1, 1, n-2, \dots$ (edges: $u+v \in \{n-1, n\}$, $u < 4s \leq 8s \leq v$); every other vertex is a leaf attached to it. *Graceful*: the differences are $n-1, \dots, 4s+1$ on the spine, $1, \dots, 3s$ on the leaves $4s..7s-1$ (attached to $4s-1$), and $3s+1+\tau(q)$ on the leaf $7s+q$, which runs over $3s+1, \dots, 4s$ exactly once. The proof in Lean (`fam_adj` characterizes adjacency, then existence and uniqueness of the edge of each difference d) uses `omega`. $\square$

Lemma 3 (`catOf_inj`, `edgeSet_ge`, `pair_ge`, `class_ge`, `complement_ge`). *For $s \geq 1$, each of the four counts at $n = 12s$ is at least $s!$.*

Proof. $\tau \mapsto G_\tau$ is injective (the neighbour of the leaf $7s+q$ among the high vertices is $10s+1+q+\tau(q)$), giving $s!$ graceful edge sets: this is (C). For (B) use (G_τ, id) . For (A), choose an isomorphism from the representative of the class of G_τ to G_τ ; the map $\tau \mapsto (\text{class}, \text{labeling})$ is injective because pulling the representative back along the labeling recovers G_τ . For (D) the class of G_τ is $\{G_\tau, \overline{G_\tau}\}$ and $\overline{G_\tau} \neq G_{\tau'}$ for all τ, τ' : in $\overline{G_\tau}$ vertex 0 is adjacent to $n-2$ (as $1 \sim n-1$ in G_τ), while in $G_{\tau'}$ it is not. So distinct τ give distinct classes. $\square$

Lemma 4 (`not_rational_of_fact`). *If $a_{12s} \geq s!$ for all $s \geq 1$, the series is not rational.*

Proof. Otherwise $s! \leq a_{12s} \leq M(R^{12})^s$ for all s , contradicting $x^s/s! \rightarrow 0$ for $x = R^{12}$. $\square$

Lean

The main theorem is `C3968.main : C3968.Claim`, where `Claim` states that none of `classCount`, `pairCount`, `edgeSetCount`, `complementCount` has a rational generating function. The statement block is at the top of `lean/Conjecture3968/Basic.lean`. Axiom audit: `#print axioms` `C3968.main` lists only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`, no `native_decide`. The definition of a graceful labeling follows the standard one (an injection of the vertices into $\{0, \dots, |E|\}$ inducing a bijection of the edges onto $\{1, \dots, |E|\}$); for a tree with n vertices and labels in $\{0..n-1\}$ this is the formalized condition. A Python check of the family for $s \leq 3$ is in the attempt record.